Decision-Centric Probabilistic Forecasting for Inventory Optimization

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Outline

Table of Contents

Point Forecasts

Point forecasts predict expected demand: $\hat{D}_{i,\tau} \approx \mathbb{E}[D_{i,\tau} | x_{i,t}]$, where:

- i = SKU / product
- τ = future time period being forecasted
- t = decision time / forecast origin
- $D_{i,\tau}$ = random demand for SKU i at time τ
- $x_{i,t}$ = feature vector available at time t for forecasting demand in period τ. Example:

$$x_{i,t} = \begin{bmatrix} \text{day of week}_\tau \\ \text{holiday flag}_\tau \\ \text{price}_{i,\tau} \\ \text{promotion}_{i,\tau} \\ \text{demand lags up to } t \\ \text{rolling demand features up to } t \\ \text{SKU / category / store attributes} \\ \vdots \end{bmatrix}$$

But inventory decisions depend on **risk**, not only expected demand.

The Newsvendor Problem I

- The Newsvendor Problem is a mathematical model used to determine the optimal inventory level for perishable or single-period goods.
- It balances the cost of overstocking (unsold items) against the cost of understocking (lost profit from unmet demand) to minimize expected total cost.
- The cost of over-forecasting and under-forecasting is asymmetric:

$$C(q, D) = h(q - D)^+ + b(D - q)^+$$

  where h is overage / holding / spoilage cost and b is underage / lost-sales / service cost.
- The optimal inventory is a quantile, not the mean:

$$q^* = F^{-1}\left(\frac{b}{b + h}\right)$$

  where F is cumulative distribution function (cdf) of the demand.

- For high service levels (95%-99%), the right tail of the demand distribution is crucial for determining the optimal inventory level.
- Two SKUs can have the same mean but very different safety stock requirements:

$$\mathbb{E}[D_1] = \mathbb{E}[D_2], \quad F_1^{-1}(0.95) \neq F_2^{-1}(0.95)$$

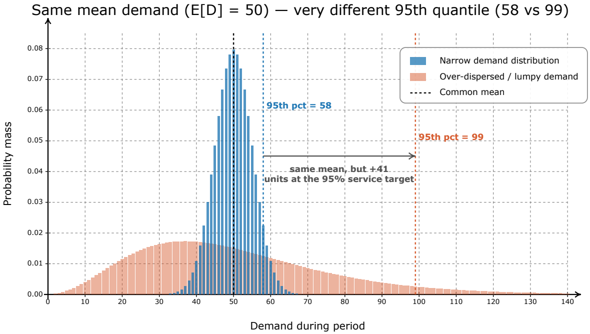


Table of Contents

Generalized Linear Models

A Generalized Linear Model (GLM)[1] is a generalization of linear regression that models the conditional distribution of a response variable, Y, given a vector of features, X.

- **Exponential Family:** The conditional distribution of the response variable Y, given X, usually belongs to the exponential family, for example Poisson or Negative Binomial. Its conditional mean is denoted as $\mathbb{E}[Y|X] = \mu$.
- **Link Function:** A monotonic, differentiable function $g(\cdot)$ that bridges the linear predictor to the expected value of the distribution: $g(\mu) = X\beta$.
- **Output:** A customized probability distribution tailored to that specific input's features, X.
- GLMs don't use ordinary least squares (OLS) like linear regression. Instead, they rely on maximum likelihood estimation (MLE).

[1]Original formulation by Nelder and Wedderburn 1972.

Poisson Regression I

- **Probability Mass Function:** Models the probability of observing y units of demand given a mean rate μ: $\mathbb{P}(Y = y \mid X) = \frac{e^{-\mu}\mu^y}{y!}$
- **Canonical Link:** Uses the log link function $\ln(\mu) = X\beta$, guaranteeing strictly positive demand predictions $\mu = e^{X\beta}$.
- **Equidistribution Property:** For Poisson distribution, the mean equals the variance: $\mathbb{E}(Y) = \text{Var}(Y) = \mu$
- **Likelihood Function:** The joint probability of n independent demand observations:

$$L(\beta) = \prod_{i=1}^{n} \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}$$

Poisson Regression II

- **Log-Likelihood Objective:** Taking the natural log yields the function maximized during optimization:

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i(X_i\beta) - e^{X_i\beta} - \ln(y_i!) \right]$$

- **Parameter Estimation:** Solved using Iteratively Reweighted Least Squares (IRLS) since no closed-form solution exists.

Negative Binomial Regression I

- **Poisson Failure:** Real-world inventory data exhibits overdispersion (Var(Y) $> \mathbb{E}(Y)$) due to bulk buying, promotion spikes, or unobserved stockouts.
- **Underestimated Risks:** Applying Poisson to overdispersed data underestimates standard errors, leading to dangerously low safety stock levels.
- **Probability Mass Function:** The Negative Binomial distribution is parameterized by:

$$\mathbb{P}(Y = y \mid \mu, \alpha) = \frac{\Gamma(y + \alpha^{-1})}{y!\,\Gamma(\alpha^{-1})} \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu} \right)^{\alpha^{-1}} \left(\frac{\mu}{\alpha^{-1} + \mu} \right)^{y}$$

- **Flexible Variance:** The relationship between mean and variance can be expressed using the dispersion parameter α: $\text{Var}(Y) = \mu + \alpha\mu^2$

Negative Binomial Regression II

- **Log-Likelihood Objective:** The optimization function used to simultaneously estimate regression coefficients β and dispersion α:

$$\ell(\beta, \alpha) = \sum_{i=1}^{n} \Big[\ln \Gamma(y_i + \alpha^{-1}) - \ln \Gamma(\alpha^{-1}) - \ln(y_i!) - \alpha^{-1} \ln(1 + \alpha e^{X_i \beta}) \\ + y_i \ln(\alpha e^{X_i \beta}) - y_i \ln(1 + \alpha e^{X_i \beta}) \Big]$$

- **Parameter Estimation:** Solved using L-BFGS.
- In Python, the easiest way to fit GLMs is using the `statsmodels` package.

Generalized Additive Models for Location, Scale, and Shape (GAMLSS)

GAMLSS[2] is a parametric framework that models every parameter of a distribution - not just the mean.

- GAMLSS allows independent modeling of up to four distribution parameters: Location (μ), Scale (σ), Shape 1 (ν, skewness), and Shape 2 (τ, kurtosis).
- Each distribution parameter can be linked to its own unique set of predictors and covariates via specific link functions.
- The framework is not locked into the exponential family.
- Ulrich et al. (2021) used GAMLSS to arrive at the cost-minimising solution to the newsvendor model in an e-grocery setting.

[2]Introduced by Rigby and Stasinopoulos 2005.

Quantile Regression I

- The **pinball loss function**[3] measures the accuracy of models that predict specific percentiles of the distribution rather than the average.

$$L_q(y, \hat{y}) = \begin{cases} q(y - \hat{y}) & \text{if } y \geq \hat{y} \quad \text{(Underestimation)} \\ (1 - q)(\hat{y} - y) & \text{if } y < \hat{y} \quad \text{(Overestimation)} \end{cases}$$

- **Model Agnostic Architecture:** Quantile regression is not limited to linear models. The objective function can be optimized using CatBoost, XGBoost, Deep Neural Networks, or any gradient-based learner.
- **Multi-Quantile Regression:** To get an accurate picture of the conditional distribution, we train models to simultaneously predict a full set of quantiles (e.g., 0.1, 0.2, ..., 0.9, 0.95, 0.99).
  - We don't need to train a separate model for each quantile.
  - Instead of producing one number, the model produces a vector of K numbers, where K is the number of target quantiles.

Quantile Regression II

- This objective function is the summed pinball loss evaluated across all specified quantiles simultaneously:

$$L_{\text{MultiQuantile}} = \sum_{k=1}^{K} L_{q_k}(y, \hat{y}_{q_k})$$

- The Cumulative Distribution Function (CDF) and the Probability Density Function (PDF) can be reconstructed from the grid of quantiles without parametric assumptions.

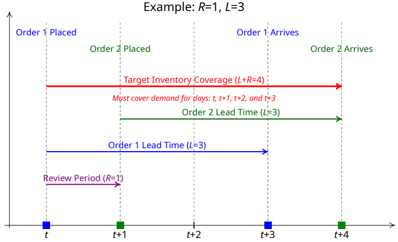
[3]First proposed by Koenker and Bassett 1978.

Table of Contents

Periodic Review Inventory Policy with Lead Time

Many real-world inventory management systems follow a periodic review policy:

- **Reorder interval** (R): The number of time periods between two consecutive orders.
- **Lead time** (L): The number of time periods an order takes to arrive.
- To avoid stockouts, the target inventory at period t must cover the cumulative demand over the $L + R$ exposure window[4]: $D_t, D_{t+1}, \ldots, D_{t+L+R-1}$.



[4]See for example "Stochastic Inventory Models: Periodic Review" 2019.

Distribution of Demand during Exposure Window $(L + R)$

The Newsvendor model can be extended to accommodate positive lead times and periodic reviews.

- **Key Idea**: Optimize for the total demand during the exposure period $(L + R)$, rather than a single period:

$$D_t^{L+R} = D_t + D_{t+1} + \cdots + D_{t+L+R-1}$$

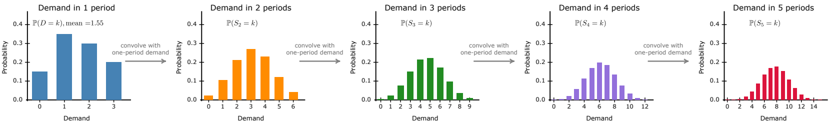
- Assuming independent periods, the probability distribution of D_t^{L+R} is computed by convolving the individual demand distributions.
- If the lead time (L) is stochastic, the distribution of D_t^{L+R} becomes a mixture distribution. It weights the cumulative demand $(S_{t,l+R} = D_t + \cdots + D_{t+l+R-1})$ across all possible lead times up to a maximum $\bar{L}$:

$$\mathbb{P}(D_t^{L+R} = k) = \sum_{l=1}^{\bar{L}} \mathbb{P}(S_{t,l+R} = k)\mathbb{P}(L = l)$$

Example: Convolution-based computation of Demand during Lead Time + Review Period

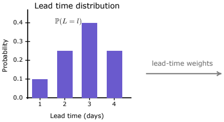
Precompute horizon-demand distributions, then mix them using the lead-time distribution.

Step 1: Repeated convolution of the one-period demand distribution
Compute and store $\mathbb{P}(S_h = k)$ up to $h =$ Max Lead Time $(\bar{L})$ + Review Period (R)



Step 2: Compute the distribution of demand during lead time + review period (D^{L+R}).

precomputed horizon distributions



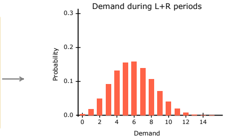
For each realized lead time l, we have
$\mathbb{P}(D^{L+R} = k \mid L = l) = \mathbb{P}(S_{l+R} = k)$

Therefore, by conditional probability:

$$\mathbb{P}(D^{L+R} = k) = \sum_{l=1}^{\bar{L}} \mathbb{P}(S_{l+R} = k)\mathbb{P}(L = l)$$

Here: $R = 1$, L takes values 1, 2, 3, 4, so horizons 2, 3, 4, 5 are mixed.

mixture / compound distribution



Example distributions used only for illustration: one-period demand support {0,1,2,3}; lead time support {1,2,3,4}.

Table of Contents

Multi-Product Newsvendor with Budget Constraint

- A decision-maker needs to determine the target inventory for N distinct products while satisfying a working capital constraint[5].
- The total capital tied up in the target inventory cannot exceed a predefined working capital limit, denoted as B.
- Inputs:
  - Demand during lead time + review period (D_i): Random variable with CDF $F_i(\cdot)$
  - Economic parameters for product i:
    - Selling price of product i: p_i per unit
    - Purchase cost of product i: c_i per unit
    - Shortage penalty of product i: r_i per unit.
  - Available budget: B

[5]Zhang, Xu, and Hua (2009) considered a similar formulation. Perakis, Singhvi, and Spantidakis (2020) further extended this to multi-item, multi-warehouse settings under dual capacity constraints.

Problem Formulation

- The profit from the sales of product i, denoted as $P_i(x_i)$, balances revenue against purchase costs and shortage penalties for the target inventory x_i:

$$P_i(x_i) = p_i \min\{D_i, x_i\} - c_i x_i - r_i \max\{D_i - x_i, 0\}$$

- Goal: Maximize the total expected profit across all N products:

$$P(x_1, \cdots, x_N) = \sum_{i=1}^{N} \mathbb{E}[P_i(x_i, D_i)]$$

- The Constraint: The total working capital required to fund the target inventory must not exceed the available limit B:

$$\sum_{i=1}^{N} c_i x_i \leq B$$

Solving via Binary Search I

- We relax the hard working capital constraint by introducing a non-negative Lagrange multiplier, $\lambda \geq 0$. This translates the constrained problem into an unconstrained maximization:

$$\text{Maximize } \mathcal{L}(x_1, \cdots, x_N, \lambda) = \sum_{i=1}^{N} \mathbb{E}[P_i(x_i, D_i)] - \lambda \left(\sum_{i=1}^{N} c_i x_i - B \right)$$

- This is equivalent to a standard newsvendor problem where the effective cost is adjusted to $c_i' = c_i + \lambda$. The optimal $x_i^*(\lambda)$ is given by:

$$x_i^*(\lambda) = F_i^{-1} \left(\frac{p_i - c_i(1 + \lambda) + r_i}{p_i + r_i} \right)$$

- Consequently, the total working capital tied up by this optimal solution, $C(\lambda) = \sum_i^N c_i x_i^*(\lambda)$, acts as a non-increasing function of λ.

Solving via Binary Search II

- Our objective is to find the smallest value of $\lambda \geq 0$ such that the total cost $C(\lambda)$ satisfies the working capital constraint $C(\lambda) \leq B$.
- Because $C(\lambda)$ is monotonic with respect to λ, we can efficiently locate the optimal λ using a binary search.
- The Procedure:
  - First, check the unconstrained solution ($\lambda = 0$). If $C(0) \leq B$, then $x_i^*(0)$ is the optimal order quantity for all products.
  - If the budget is exceeded, initiate a binary search over the interval $[0, \lambda_{\max}]$, where $\lambda_{\max} = \min_{i=1,\cdots,N}\left(\frac{p_i+r_i}{c_i} - 1\right)$.
  - Set $\lambda_L = 0$, $\lambda_U = \lambda_{\max}$, and evaluate the midpoint $\lambda_M = (\lambda_U + \lambda_L)/2$.
  - If $C(\lambda_M) > B$, then λ_M is too small; continue the search in $[\lambda_M, \lambda_U]$.
  - If $C(\lambda_M) \leq B$, then λ_M is potentially feasible but might be too large; continue the search in $[\lambda_L, \lambda_M]$.

Table of Contents

Takeaways

- Forecasts are not the objective, inventory decisions are.
- Probabilistic forecasting estimates the entire conditional demand distribution: $\mathbb{P}(D_{t+\tau} \mid x_t)$
- Lead time and review period uncertainty are propagated by computing the distribution of cumulative demand:

$$D_t^{L+R} = D_{t+1} + \cdots + D_{t+L+R}$$

- Inventory targets are obtained by solving a stochastic optimization problem that explicitly accounts for
  - service level requirements,
  - holding and shortage costs,
  - and working capital constraints.

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